

Question 1 – Real-World Scene Capture to Digital Image Representation

1. Real-World Scene

- A three-dimensional scene contains objects with different shapes, colors, and surface properties.

2. Illumination Source

- A natural or artificial light source illuminates the scene.
- The quality and intensity of illumination affect image appearance.

3. Surface Reflectance

- Object surfaces reflect light based on their material properties.
- Reflection may be diffuse (Lambertian) or specular (mirror-like).

4. Optical Lens System

- The camera lens collects and focuses the reflected light onto the image sensor.
- This forms a continuous image on the imaging plane.

5. Continuous Image Formation

- The scene is represented as a continuous two-dimensional image function $f(x, y)$.
- Image intensity depends on both illumination and surface reflectance.

6. Spatial Sampling

- The CMOS/CCD sensor samples the continuous image into discrete pixels.
- This determines the spatial resolution of the digital image.

7. Amplitude Quantization

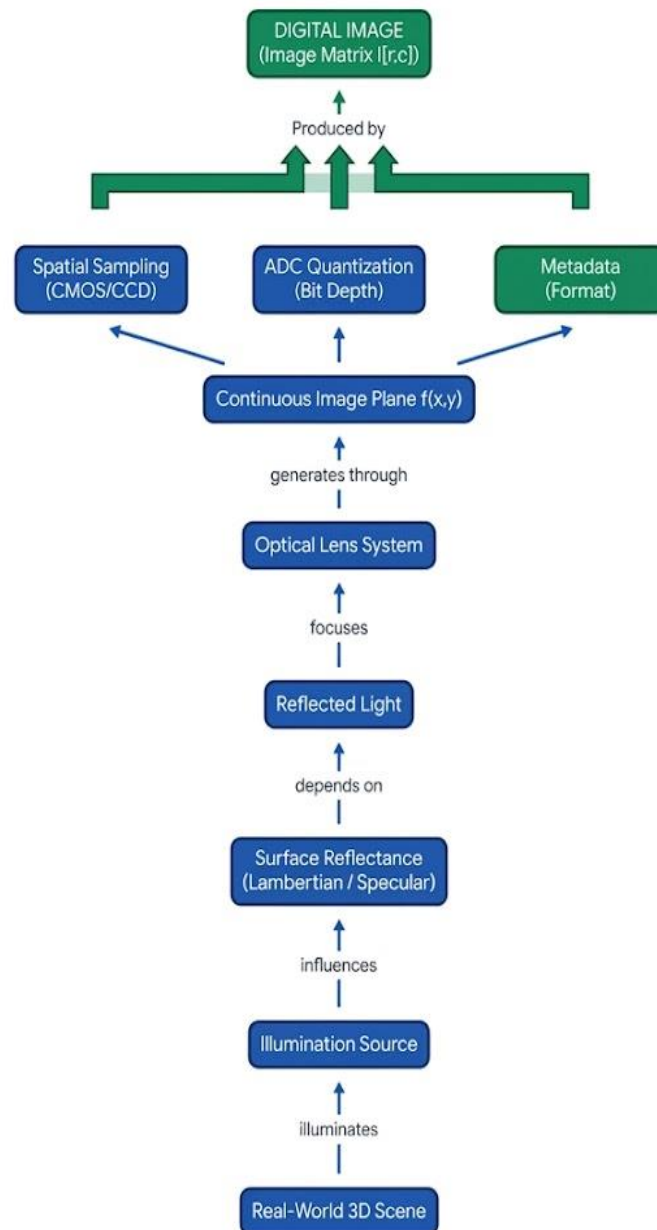
- An Analog-to-Digital Converter (ADC) converts the continuous pixel intensity into discrete digital values.

8. Digital Image Representation

- The final output is a digital image matrix $I(r, c)$.
- Each pixel stores a numerical intensity value that can be processed, analyzed, and interpreted by computer vision systems.

Concept Map Diagram

Q2: Re-ordered Physical Image Formation to Digital Data



Question 2 – Computer Vision Data Handling Pipeline

1. Image Acquisition

- Capture images or videos using a camera, sensor, or imaging device.
- Convert the real-world scene into digital image data.

2. Image Preprocessing

- Improve image quality before analysis.
- Perform operations such as noise removal, contrast enhancement, resizing, normalization, and color correction.

3. Feature Extraction

- Identify important visual characteristics from the image.
- Extract features such as edges, corners, textures, shapes, keypoints, or deep learning features.

4. Feature Analysis

- Analyze the extracted features using Computer Vision algorithms.
- Perform tasks such as image classification, object detection, image segmentation, object tracking, or pattern recognition.

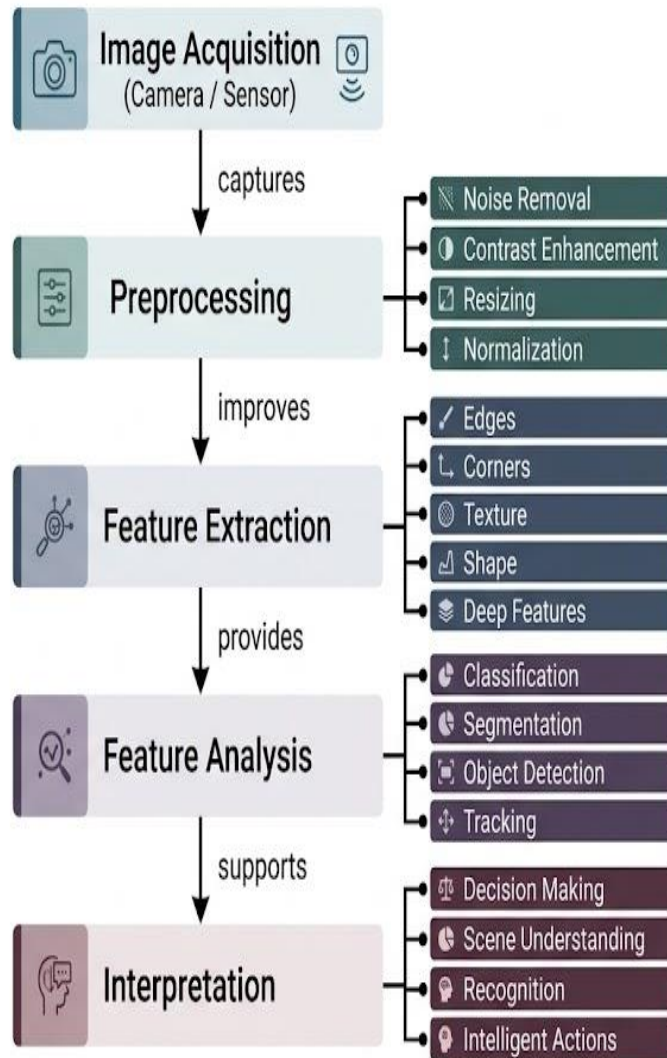
5. Interpretation and Decision Making

- Convert analysis results into meaningful information.
- Recognize objects, understand scenes, make predictions, and support intelligent decision-making for real-world applications.

6. Application Output

- Generate the final output based on the interpreted results.
- Applications include facial recognition, autonomous vehicles, medical image analysis, surveillance, industrial inspection, and robotics.

Concept Map Diagram



Question 3 – Relationship among Resolution, Spatial Resolution, Intensity Resolution and Image Quality

1. Image Resolution

- Image resolution refers to the total number of pixels in a digital image (width × height).
- Higher image resolution captures more visual detail and improves image clarity.

2. Spatial Resolution

- Spatial resolution represents the size and density of pixels in an image.
- Smaller and more closely packed pixels produce sharper and more detailed images.

3. Intensity Resolution (Bit Depth)

- Intensity resolution indicates the number of brightness or color levels available for each pixel.
- Higher bit depth provides smoother intensity transitions and better color representation.

4. Image Quality

- Image quality depends on the combination of image resolution, spatial resolution, intensity resolution, and noise level.
- Better image quality improves visibility, sharpness, and feature recognition.

5. Impact on Applications

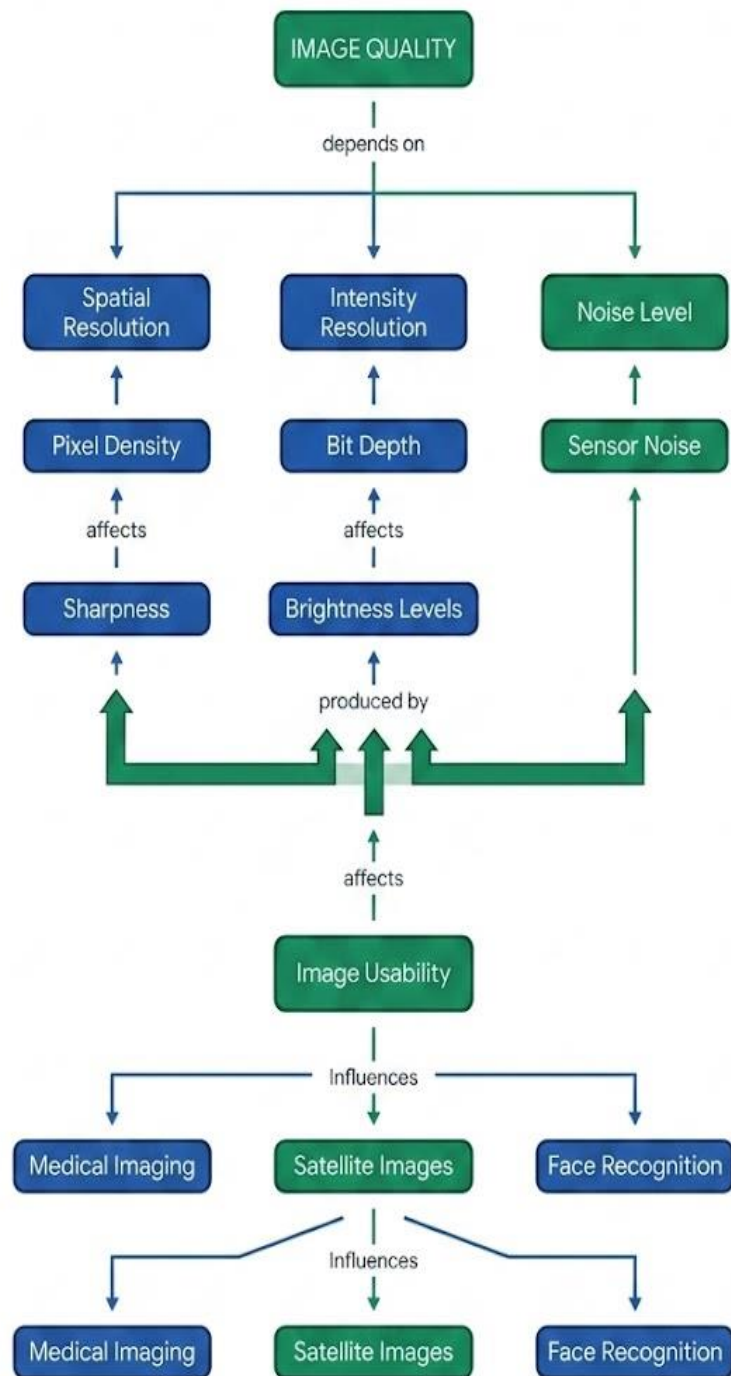
- Medical Imaging: High resolution enables accurate diagnosis.
- Satellite Imaging: Fine spatial resolution improves terrain and object identification.
- Face Recognition: Better image quality increases recognition accuracy.
- Industrial Inspection: High-quality images help detect defects more precisely.

6. Relationship Among the Factors

- Image resolution, spatial resolution, and intensity resolution are interdependent.
- A balanced combination of these factors produces high-quality images suitable for accurate analysis.

Concept Map Diagram

Q3: Understanding Image Quality and its Impact



Question 4 – Factors Influencing Image Acquisition Conditions

1. Illumination

- The brightness, direction, and color of the light source affect image appearance.
- Poor lighting can cause shadows, glare, and low contrast, reducing image quality.

2. Sensor Characteristics

- Image sensors (CMOS or CCD) convert light into digital signals.
- Sensor resolution, sensitivity, dynamic range, and noise level determine the quality of the captured image.

3. Motion

- Camera movement or object movement during image capture may produce motion blur.
- Stable cameras and faster shutter speeds help reduce blur and improve image sharpness.

4. Environmental Conditions

- Weather conditions such as fog, rain, dust, and smoke can reduce image visibility.
- Temperature and humidity may also affect camera performance and image clarity.

5. Optical System

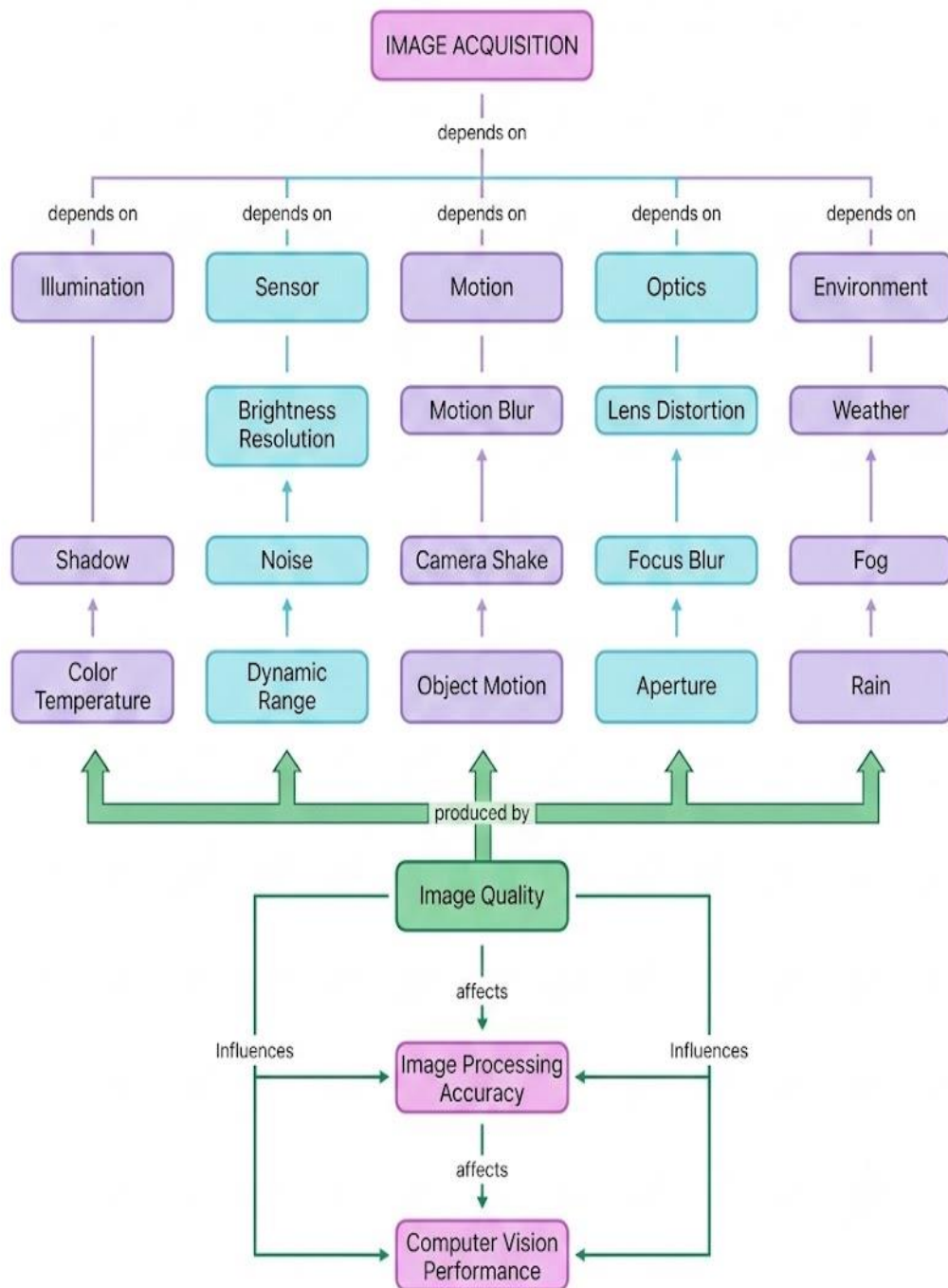
- Lens quality, focal length, aperture, and focus determine image sharpness and field of view.
- Optical distortions and poor focus can reduce image accuracy.

6. Image Acquisition Quality

- Proper control of illumination, sensors, motion, environment, and optics produces high-quality images.
- High-quality images improve the reliability of subsequent image processing tasks.

Concept Map Diagram

Q4: Analyzing Image Acquisition and its Downstream Impacts



Question 5: Relationship Between Computer Vision System Capabilities and Vision-Based Applications

1. Image Understanding

- The system captures and interprets visual information from images or videos.
- It forms the foundation for all Computer Vision tasks.

2. Object Detection

- Identifies and locates objects within an image.
- Widely used in surveillance, traffic monitoring, and autonomous vehicles.

3. Object Recognition

- Classifies and identifies detected objects based on learned features.
- Used in facial recognition, biometric authentication, and retail systems.

4. Image Segmentation

- Divides an image into meaningful regions or objects.
- Supports medical image analysis, autonomous driving, and satellite image interpretation.

5. Object Tracking

- Continuously follows the movement of objects across video frames.
- Applied in video surveillance, sports analysis, and robotics.

6. Measurement and Inspection

- Measures object size, shape, distance, and detects defects.
- Commonly used in industrial quality control and manufacturing.

7. Decision Making

- Combines outputs from different Computer Vision capabilities to make intelligent decisions.
- Improves automation, accuracy, and efficiency in complex tasks.

8. Vision-Based Applications

- Computer Vision capabilities are applied in healthcare, autonomous vehicles, surveillance, agriculture, robotics, industrial inspection, retail, smart cities, and remote sensing.

Concept Map Diagram

